Tutorial 3

Link Analysis

Problem 1

$$\begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \end{pmatrix} \end{array}$$

C is a dead-end, so

$$M = \begin{pmatrix} 0 & 1/2 & 1/3 \\ 1 & 0 & 1/3 \\ 0 & 1/2 & 1/3 \end{pmatrix}$$

Problem 2

$\beta = \frac{4}{5}$

a) $$M = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 0 & 1/2 \\ 1 & 0 & 0 \\ 0 & 1 & 1/2 \end{pmatrix} \end{array}$$

If (a,b,c) are PageRank for (A,B,C), then

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \frac{4}{5}\begin{pmatrix} 0 & 0 & 1/2 \\ 1 & 0 & 0 \\ 0 & 1 & 1/2 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} + \frac{1}{15}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$a = \frac{2}{5}c + \frac{1}{15}$$

$$b = \frac{4}{5}a + \frac{1}{15}$$

$$c = \frac{4}{5}b + \frac{2}{5}c + \frac{1}{15}$$

$$\Rightarrow \quad a = 0.25581395, \quad b = 0.27131783, \quad c = 0.47286822$$

b) $$M = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 0 & 1/2 \\ 1 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix} \end{array} \Rightarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \frac{4}{5}\begin{pmatrix} 0 & 0 & 1/2 \\ 1 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} + \frac{1}{15}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$a = \frac{2}{5}c + \frac{1}{15}$$

$$b = \frac{4}{5}a + \frac{2}{5}c + \frac{1}{15}$$

$$c = \frac{4}{5}b + \frac{1}{15}$$

$$\Rightarrow \quad a \approx 0.22012579, \quad b \approx 0.39622642, \quad c \approx 0.3836478$$

Naïve Bayesian Classification

Problem 1 $X = (A=1; B=2; C=2)$

$$P(\text{Class}=1 \mid X) = \frac{P(X \mid Cl=1) \cdot P(Cl=1)}{P(X)}$$

$$P(Cl=2 \mid X) = \frac{P(X \mid Cl=2) \cdot P(Cl=2)}{P(X)}$$

$$P(X \mid Cl=1) \cdot P(Cl=1) = P(A=1 \mid Cl=1) \cdot P(B=2 \mid Cl=1) \cdot P(C=2 \mid Cl=1) \cdot P(Cl=1)$$

$$= \frac{2}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{4}{7} = \frac{1}{8 \cdot 7} = \frac{1}{56}$$

$$P(X \mid Cl=2) P(Cl=2) = P(A=1 \mid Cl=2) \cdot P(B=2 \mid Cl=2) \cdot P(C=2 \mid Cl=2) \cdot P(Cl=2)$$

$$= \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{3}{7} = \frac{4}{63}$$

$\frac{4}{63} > \frac{1}{57}$ => Answer: $\text{class}(X) = 2$

Problem 2: Similar to the Problem 1.